

MIKHAIL EMAD ALBER

Software Engineer | Full-Stack & Systems Architecture

Cairo, Egypt

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GitHub

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PROFESSIONAL SUMMARY

Results-oriented Software Engineer with extensive experience architecting and deploying scalable web services, high-throughput REST APIs, and automated data pipelines. Delivered 40+ engineered applications leveraging Python (Django, Flask), JavaScript (React, Node.js), relational & NoSQL databases (PostgreSQL, SQL Server, MongoDB), and Linux server infrastructure. Proven expertise in microservices architecture, JWT/RBAC security, real-time WebSockets, database query optimization, and production environments maintaining high availability.

TECHNICAL SKILLS

Languages: Python, JavaScript (ES6+), TypeScript, SQL (PostgreSQL, MySQL), HTML5, CSS3, Bash Shell

Backend & APIs: Django, Django REST Framework (DRF), Flask, Node.js, Express, RESTful APIs, WebSockets, JWT Authentication, Microservices

Frontend & UI: React.js, Next.js, Redux, State Management, Responsive Web Design, Bootstrap, Tailwind CSS, Three.js / WebGL

Databases & Storage: PostgreSQL, Microsoft SQL Server, MySQL, MongoDB, Redis, Schema Design, Indexing, Query Optimization

DevOps & Infrastructure: Linux / Ubuntu Server Administration, Docker, Git, GitHub, CI/CD, Nginx, cPanel, SiteGround Hosting, Reverse Proxy

AI & Data Engineering: ETL Pipelines, Data Analysis, Machine Learning, Deep Learning, OpenCV, TensorFlow, Keras, CNNs, NLP, Transformers, Pandas

Methodologies & Core: System Design, Object-Oriented Programming (OOP), Data Structures & Algorithms, Agile / Scrum, Root-Cause Analysis

WORK EXPERIENCE

Systems & Software Support Engineer | **Petra Software**

2025 - Present | Cairo, Egypt

- Administer and maintain production Ubuntu Linux server environments across SiteGround and cPanel, sustaining high application uptime and reliability for enterprise client portals.
- Deploy and configure full-stack web applications on live cloud instances, implementing automated verification checks, SSL/DNS configurations, and continuous release rollouts.
- Isolate and resolve tier-2/3 system defects, database deadlocks, and API bottlenecks, coordinating with core development teams to accelerate issue resolution.
- Optimize SQL database configurations, query performance, and server resource utilization, establishing standard operating procedures and technical troubleshooting runbooks.

TECHNICAL PROJECTS

Enterprise Restaurant POS & Operations Platform

React | Django REST | PostgreSQL | WebSockets | JWT

- Engineered high-concurrency order dispatch utilizing real-time WebSockets, decreasing order transmission latency to under 250ms across 50+ concurrent POS terminals.
- Automated multi-station kitchen display routing and network thermal printing across 3 prep stations, reducing order fulfillment transit times by 45%.

Hospital Clinical & Operations Management System

React | Django REST | PostgreSQL | RBAC Security

- Architected enterprise healthcare suite coordinating electronic health records (EHR), outpatient scheduling, and financial ledgers across 8 hospital departments.
- Automated real-time pharmacy inventory deduction upon prescription issuance, eliminating dispensing errors and accelerating cross-department billing audits by 50%.

Retail Data Warehouse & Analytics Platform

Python | SQL Server | MongoDB | Flask | Pandas | ML

- Constructed an end-to-end ETL data pipeline ingesting 100,000+ relational transactions from SQL Server into MongoDB, exposing aggregated analytics via sub-100ms Flask APIs.
- Integrated predictive machine learning algorithms for sales analysis and inventory demand forecasting, reducing stockout risks by 28% and optimizing supply chain reorders.

Real-Time Edge Computer Vision & Detection Engine

Python | OpenCV | TensorFlow / Keras | MobileNet CNNs

- Built an edge-optimized computer vision inference pipeline simultaneously evaluating mask compliance and classifying multi-class emotional states on live video streams with 94%+ accuracy at 30+ FPS.

EDUCATION

Bachelor of Science in Computer Science and Engineering | **The Egyptian E-Learning University (EELU)**

Expected 2025

CERTIFICATIONS

Data Science & Artificial Intelligence Specialization | **AMIT Learning**

2026

Full-Stack Web Development Specialization | **AMIT Learning**

2024 - 2025

Artificial Intelligence & Cloud Computing Training | **Huawei Technologies**

2022

LANGUAGES

English: Professional Working Proficiency (B2) | **Arabic:** Native Proficiency